

# Structure, new materials, and new manufacturing technologies

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## 1 Introduction

The main objective of the structural system is to mechanically support and protect all spacecraft subsystems during the different mission phases and, in the case of CubeSats, to provide a mechanical interface to the deployer system. From the manufacturing phase to the end of the mission, the satellite faces different environments and different loads. The structure should guarantee the correct interface with the launch vehicle that, in the case of CubeSats, is partially determined by the deployer selected (see [Chapter 22](#)). The CubeSat standard specification document [1,2] gives a clear description of the constraints that the satellite should address, in order to be considered a CubeSat. Nevertheless, some of the constraints that dictate the structural design are derived from the mission and systems requirements. For this reason, it is important to give the CubeSat developer some useful tips and suggestions that can help them select the best structure for their mission.

To result in a successful structural product, the process can be divided into five steps: (1) requirements definition, (2) design or selection process, (3) analysis, (4) assembly, and (5) verification and testing. In the case of the CubeSat, some phases are standardized, resulting in a simplification of the work that is required of the satellite developer, but others are not. In addition, some additional aspects and restrictions with respect to traditional satellites need to be considered, such as the very small dimensions required, often dictate the use of deployable components.

Traditionally, the design of space systems makes a distinction between the *primary structure*, defined as the one that carries the major loads, and the *secondary structure*, defined as the one that supports components under 5 kg. When considering CubeSats, in particular 1U structures, this distinction does not make sense in terms of the weight to be considered but can still be meaningful in terms of the function of the structure. CubeSat primary structures are the ones designed to transmit loads through the spacecraft to the interface of the launch and deployment system and the ones that serve as the mechanical interface with the other bus systems, the payloads, and their associated components. The function of the secondary structures, on the other hand, is only to support themselves. Their failure does not necessarily result in a catastrophic event as in the case of a primary structure failure. A typical example of primary structure is the satellite chassis, while an example of a secondary structure is the antenna system.

In the process of designing and selecting the best structure that matches the mission and subsystem needs, structural engineers are strongly supported by the commercial availability of different solutions. An interesting report from NASA gives an overview of all the different primary structures and mechanisms available on the commercial market [3]. In some cases one of them may be the perfect match for the particular mission that will be performed, while in other cases a customized solution is needed.

This chapter gives insight into how the developers should orientate themselves to optimize the design, giving suggestions and indications coming from practical experiences. The chapter follows a step-by-step approach to help the CubeSat developer: from requirements definition to the design and analysis, manufacturing, and eventually testing for verification and validation.

## 2 Requirements and main characteristics

During the system design process, the first step that should be completed is the requirements definition. This step defines what the structure should perform and what its main goals are. A clear, accurate, and detailed requirements definition allows the designers to start an optimized process where time and costs are reduced to the minimum. If the developers know exactly what requirements the structure should satisfy, from the beginning of the project, they will be able to focus their attention clearly on the development of the best solutions for that specific mission.

Some conditions and constraints can change during the project development. As an example, the loads experienced at launch significantly affect the structural design. If, for some reasons (political, availability, price changes, etc.), the launch vehicle changes, the requirements can change dramatically, potentially causing the need of replacing the structure selected, affecting the overall costs and schedule.

A clear definition of the requirements is not always possible during the initial mission analysis phases, but adopting a conservative and iterative approach can help minimizing the effect of last minute changes. A conservative approach is adopted when considering the launch loads, for example.

Generally speaking, the structure should be:

- **rigid**, or able to support the dynamics loads during all the mission phases;
- **robust**, or able to support the quasi-static loads during the mission;
- **light**, to reduce the launch costs;
- **accessible**, in terms of easiness of access to the main subsystems if and when needed;
- **easy** to manufacture and to assemble.

When defining the structure requirements, developers need to concentrate their attention in two different areas:

- **internal requirements**: the ones that are dictated by the mission and the interface with the other subsystems;
- **external requirements**: the ones related to the launch vehicle, the mission phases, and the space environment.

Each of these is considered in the following text.

## **2.1 Internal requirements**

The structural internal requirements can be divided in two main categories:

- requirements dictated by the mission;
- requirements dictated by the bus.

These vary in function, being unique to the mission, but there are some general considerations that can be analyzed.

### **2.1.1 Requirements dictated by the mission**

The main goal of the structural system is to be able to accommodate and support the payload during all the mission phases. Requirements on the structure are not limited to dimensions compatibility with the payload, but will be dictated by the payload functions too. In an imaging satellite, for example, it is critical that the payload is accommodated in such a way that its aperture will have access to the exterior of the satellite. In other cases, this requirement might not be well defined. An example is the case of biomedical payloads, that usually require three levels of containment. Whether this is needed also in a small satellite mission depends on several other factors that should be analyzed during the preliminary mission design.

To support the payload during all the mission phases, the structure should support loads and operational conditions that depend on the mission nature and configuration. The resulting requirements are strictly related, sometimes overlapping, with some concerns regarding the bus and some related to the external requirements. This is due to the fact that to support the payload, developers need to guarantee that not only the payload but also the entire satellite bus will be able to work properly in an extreme environment and under specific stressful conditions such as those faced during the launch.

### **2.1.2 Requirements dictated by the bus**

The structural system design is subject to constraints and requirements dictated by several of the primary satellite bus system requirements. Bus elements that need to be considered in the structural design include the Electric Power System (EPS), the Telecommunications, Tracking and Command (TT&C) System, the Attitude Determination and Control System (ADCS), the Orbital Determination and Control System (ODCS), the electronics in general, and the thermal management system.

#### **Electric power system**

The power requirements indirectly drive the structural design in several ways. The power produced onboard by the solar panels, for example, is strictly related to the size of the solar panels, to the packing factor, and to the choice of body mounted or deployable solar panels. These three factors drive the selection of one structure instead of another.

Battery selection, on the other hand, influences the satellite total mass, and the design should take into account the way the different EPS subsystems are connected

and how power is distributed to the other systems with the proper voltages and currents and with the proper regulation. For CubeSats, this can be simplified owing to the use of standardized solutions and connectors. For example, the concept of having the boards stacked one on top of the other (such in the case of the use of PC104 standard) or inserted into one common board that act as an integrated backplane/connector (as in the case of the AESP-14 satellite [4]) reduces the use of wires inside the satellite and limiting the difficulties related to the cabling.

### Telecommunication, tracking and command

The TT&C system influences the structural design, due to the antenna requirements. The antenna dimensions are strictly correlated with the wavelength and, therefore, the frequency used for the transmission. Due to the satellite's small dimensions, when working with CubeSats operating at UHF or VHF frequencies, deployable antennas need to be considered. In the CubeSat mission design specifications [1], it is required that "all deployables such as booms, antennas, and solar panels shall wait to deploy a minimum of 30 minutes after the CubeSat's deployment switch(es) are activated from -POD ejection." From a structural point of view, this means that a deployable mechanism, for example an antenna, should be implemented as part of the secondary satellite structure. When using higher frequencies, patch antennas can be the best solution; the areas and the positions allocated for the antenna locations need to be considered.

Another aspect that should be taken in to account is the antenna radiation pattern that can be influenced by the satellite structure. The use of specific connectors also drives the structural configuration and should be taken into account during the design. For lower-frequency antennas, having the proper orientation and sufficient area for the ground plane should also be considered.

### Attitude determination and control system

During the structural design, designers need to take into account the mass distribution of the satellite and how it will influence the inertia matrix, since it strictly constraints the ADCS system. In some cases, the selected ADCS solution imposes a specific structural configuration. For example, the UniCubeSat-GG satellite, designed to study the librations due to the gravity-gradient stabilization, was characterized by the use of two booms, and two tip masses were used to stabilize the satellite (Fig. 1). The need to have a minimum inertia axis, achieved through the implementation of the two booms, completely constrained the satellite design and the system positioning inside the 1U CubeSat. The use of deployment mechanisms affecting the ADCS system should be considered during the design phase, since these systems can cause a torque during on-orbit operations.

ADCS and structures are also connected by the sensors' and actuators' volume allocations. A star sensor, for example, needs to be installed on a side facing the sky. However, this is not only a constraint for the ADCS sensor and actuators, but it also influences the position of the other systems' components. Another example is the design of a LEO satellite equipped with a GNSS receiver and a S-band transmitter,



**Fig. 1** UniCubeSat-GG Structure with deployable booms.

for which it should be ensured that the GNSS antenna will face higher orbits, where the navigation satellites are located, and the S-band should point toward the Earth, where the ground stations are located. The position of both components can be determined only if the satellite's attitude position during the different orbit phases is clearly defined.

### Orbital determination and control system

As for the ADCS system, the ODCS is influenced by the mass distribution, in particular by the position of the Center of Gravity (CG). For example, if a thruster for orbital maneuvering is implemented in the system, it should be aligned with the position of the CG to avoid the generation of additional torques. Similar to the ADCS, sensors' and actuators' position is crucial for this system and should be carefully taken into account during the structure design.

### Electronics

When considering the structural design, one needs to remember that the system is working in the space environment and will be adversely affected by space radiation. One of the functions of the structure is to ensure the survival of satellite components during all the mission phases. This requires the structural engineer to also consider advanced solutions to protect the satellite's internal components, in particular the electronics from space radiation.

Another important point that constrains the structural design are the electronics interfaces used to assemble the different electronics boards. One of the most common standards for CubeSat electronics is the PC104 form factor. It allows developers to stack the boards one on top of the other, facilitating the connection between the different boards. The use of this standard constrains the satellite internal design, but at the same time, the use of standoffs between the boards helps to increase the structural performance of the satellite itself.

## Thermal system

The strict connection between the structural and the thermal systems often forces the developer to consider the thermal system as a subsystem of the structural system. In space, due to the environment and in particular due to the vacuum conditions, heat transfer mainly occurs through radiation and conduction, with convection only occurring in the artificial atmosphere of the manned systems. This means that the thermal system is limited by the position of the different systems and by the materials used to manufacture them. In this way, through material selection and component positioning, it is possible to constraint and modify the temperature exchange between internal components of the satellite and between the satellite and its environment.

## 2.2 External requirements

The external requirements can be divided into three main categories:

- requirements dictated by the launch vehicle;
- requirements dictated by the space environment;
- requirements dictated by the deployer.

### 2.2.1 Requirements dictated by the launch vehicle

The requirements related to the launch vehicle selection can be considered the most important for the structural design. Designing a structural system that is able to survive to the launch loads is one of the primary objectives for the structural engineer. The launch vehicle selection affects structural design mainly by requiring that the structure is rigid enough to sustain the loads during the launch. These conditions change as a function of the launch vehicle selected. It is possible to divide the loads that the structure is subject to during launch into two main categories: quasi-static and dynamical loads, as will be discussed in [Section 4.2.1](#). The approach that should be adopted is based on the following:

- the identification of the loads, as a function of the launch vehicle;
- the analysis of the load effects on the structure, using simulations;
- test and verification of the capabilities of the real structure to sustain the predicted loads.

CubeSat structures have an extreme advantage over traditional satellites: they are not constrained by the mechanical interfaces with the launch vehicle as is the case of conventional satellite systems. Their mechanical interfaces are with the deployer and do not change as a function of the launch vehicle. CubeSats are flexible and can be manifested on different launch platforms and the developers do not need to wait for a particular launch if the mission does not need a specific orbit. On the other hand, this peculiarity of CubeSats means that sometimes the structural designer is not aware of the launch vehicle selected and consequently cannot exactly identify the loads. To help during the process, some space agencies, as for example JAXA [\[5\]](#), provide specific suggestions to the loads that should be considered if the mission plans to launch from the International Space Station, considering different launch vehicles that could be used (HTV, ATV, Space X Dragon, and Orbital's Cygnus).

In addition to the traditional loads related to the launch vehicle, the loads generated during transportation to the launch site shall be considered as well. These loads are of a different nature and cannot be underestimated given that they can cause several damages to the satellite flight unit, particularly with reference to shock. To prevent damage during transportation, it is important to use protective shockproof, dustproof, and waterproof enclosures and to monitor the shock and humidity environment in these enclosures during transportation.

### 2.2.2 *Requirements dictated by the space environment*

In the process of designing a structural system, the developers need to consider where the system will operate and what its interfaces with the external environment are. In the case of low Earth Orbit (LEO) satellites, designers face an extreme environment with characteristics that deeply influence the design selections. **Vacuum conditions**, for example, impose a choice of materials that will have a reduced outgassing to reduce the total mass loss (TML) and/or the collected volatile condensable material (CVCM) under certain levels dictated by the launch vehicle or standards (see, e.g., [6]). In the case of launch from the International Space Station, for example, Rating “A” materials that are identified in [7] shall be used for a satellite. In other words, they should comply with the following low outgassing criterion per ASTM-E595-84:

- total mass loss (TML)  $\leq 1.0\%$
- collected volatile condensable material (CVCM)  $\leq 0.1\%$

Material selection is also constrained by UV degradation and degradation due to exposure to atomic oxygen, both typical of the space environment. **Atomic oxygen** oxidizes many metals, especially silver, copper, and osmium, and it reacts strongly with any material containing carbon, nitrogen, sulfur, and hydrogen bonds, meaning that many polymers react and erode. Even materials with atomic oxygen-protective coatings can degrade. **UV degradation** causes damage to polymers through either cross-linking (hardening) or chain scission (weakening). UV under high vacuum can also create oxygen vacancies in oxides, leading to significant color changes. Contrary to the atomic oxygen that may bleach materials, UV generally darkens them, particularly in the presence of contamination. Another space environment condition that should be taken into account is the presence of **plasma** that imposes grounding requirements on the spacecraft to avoid static charge buildup, sputtering, arcing, and parasitic currents. The effects of space radiation and the interface with the extreme thermal environment have previously been discussed.

### 2.2.3 *Requirements dictated by the deployer*

CubeSats do not have a direct interface to the launch vehicle but need to be installed into a deployer. An analysis of the different deployers is provided in [Chapter 22](#) of this book. In general, to comply with the constraints of the majority of commercially available CubeSat deployers, developers need to begin with the requirements defined in the CubeSat design specifications [1]. In addition to ensuring that the external satellite

envelope complies with the dimensions defined by [1], it is important to consider the position of the following:

- the access ports;
- the switches;
- the spring plungers (when needed);
- the CG.

Other CubeSats characteristics, including the use of deployment mechanisms and structures such as the antennas, and anodized rails should be designed in accordance with the standard [1]. Most of the commercially available deployers conform to the standards defined in the CubeSat design specifications document, but it is suggested that each deployer's interface control document (ICD) be referred to as a variety of deployer-related dimensions, volumes, masses, and mechanical and electrical interfaces exist nowadays.

### 3 Design and verification process

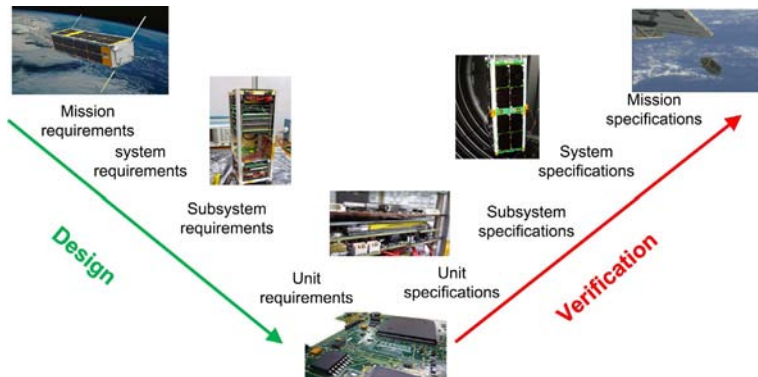
The design process is defined as the process used to generate the set of information describing the essential characteristics of a product [8]. Design means developing requirements, identifying options, doing analyses and trade studies, and defining a product in enough detail so that it can be built to specification [9].

On the other hand, the verification process is defined as confirmation by examination and provision of objective evidence that specified requirements have been fulfilled [10]. Verification means providing confidence through disciplined steps that a product will do what it is supposed to do [9].

The design and verification processes are usually described using a V-diagram as the one illustrated in Fig. 2, which represents the mission design and verification process for a specific 3U Mission (SERPENS). This is a common approach for all system design and verification processes. The following paragraphs describe in detail how designers can apply this approach to the structural design. Developers need also to remember that the verification process can be divided in two parts: the analyses that are conducted during the structural design and the tests that are conducted after the manufacturing phases. In this paragraph, only the analyses that should be conducted since they are integrated into the design process are presented.

#### 3.1 Structural design

The first step in the design phase is to have a clear idea of all the structural requirements, described in Section 1. From the analysis, it is possible to identify the structure constraints for the specific mission. How these constraints correlate with the CubeSat design specifications provides specific indications for the structure design and selection. Unlike other satellites, where a form factor is not specified, CubeSat developers have the opportunity to avoid the main frame structural design by selecting a commercially available structure. In fact, since the beginning of the CubeSat era, a number of



**Fig. 2** SERPENS mission V-diagram design and verification.

commercially available structures have been designed. The NASA report State-of-the-art in Small Spacecraft Technology [3] provides an interesting overview of the most commonly used structures that are available to date.

Effectively, CubeSat developers have three options for designing the structure:

1. selecting an existing commercially available structure, already qualified and tested;
2. developing their own structure;
3. customizing an existing structure.

The choice depends of course on the requirements and the constraints, and each one of these opportunities has advantages and disadvantages. If developers select the second opportunity, they should start a design process that depends on the experience and capabilities of the engineering team itself. The process can be resumed in the following steps:

1. requirements definition;
2. first design;
3. verification of the structure characteristics in terms of total mass, respecting the CubeSat design specifications;
4. design modifications;
5. final analysis.

The process is iterative, and the final result will be an optimized structure ready for manufacturing.

Several tools can be used during this phase, such as CAD software and FEM software. In particular, some CAD softwares offer the opportunity to perform virtual Fit-Checks not only between the satellite and the deployer but also between the different internal components. During the design process, it is extremely important to consider not only the structure itself but also how it should be assembled, what hardware works with the limited space available, and what sequence should be followed to integrate the different components. To know, for example, that during the assembly whether a specific fastener can be accessed and torqued or not, or if a connector can be reached

will be extremely important. All these aspects can be simulated in advance using CAD tools. This approach will avoid wasting time and money and will ensure that the product not only satisfies the requirements but it is also easy to assemble.

Other important issues that should be looked at are related to manufacturing capabilities. Not all complex structures can be manufactured, and it is important to consider also how much material will be wasted during the manufacturing process to constrain the costs. The use of rapid prototyping technologies such as 3-D printing can help solving several issues related to the interface between design and manufacturing capabilities, but the engineer should always keep the design simple.

## 3.2 *Structural analysis*

Structural analysis can be considered a part of the iterative design process. Based on its results the engineer can decide which changes in the design will yield better performance and an optimization of the structure. Structural analysis can usually be performed using commercially available FEM software that allows designers to perform static and dynamical analyses on the structural model. The CAD file can be directly imported into the FEM software, but, in some cases, due to the complexity of the structure, it is easier to design a simplified model for use with the FEM tools. Some CAD software systems have tools and features that can be used to also perform dynamical and static analyses.

The main steps in conducting the FEM analysis are as follows:

- importing the CAD model, remembering to apply the correct materials;
- defining the FEM model using nodes and elements;
- applying the loads that depend on the analysis that should be conducted;
- defining the constraints;
- running the analysis;
- analyzing the results.

A detailed explanation of how to perform the analysis is beyond the scope of this chapter, but it is important to remember to define the main loads that should be considered for the analysis.

### 3.2.1 *Loads*

During a space mission, several loads will be applied to the satellite structure. Mainly, they are due to the launch conditions and are well defined in the interface control manual (ICD) between the rocket and the spacecraft or in the launch vehicle manual. Other loads can be caused by opening mechanisms, transportation or by thermal loads, such as the stresses caused when two materials, with different coefficient of thermal expansions, are coupled together. In general the loads are caused by forces and accelerations and can be divided in three categories:

- **static loads**, which are time independent;
- **dynamic loads**, which are time dependent and for which inertial effects cannot be ignored;

- **quasi-static loads**, which are time dependent but are “slow” enough so that inertial effects can be ignored [11].

Note that a quasi-static load for a given structure may not be quasi-static for another structure (made of a different material). Spacecraft structures are not normally affected by static loads, but quasi-static loads caused by steady state accelerations are typical of launches. Dynamic loads can be divided into low-frequency vibrations, broadband vibrations (such as random vibrations and acoustic loads) and shocks. While the various loads have different origins and time constants, all must be considered during the design of the spacecraft structure.

## 4 Materials and manufacturing

Material selection and manufacturing technologies are key considerations in producing an optimized structure. CubeSat design specifications impose the use of aluminum alloys 6061, 7075, 5051, or 5005 for the main CubeSat structure and the rails, making material selection a comparatively easy choice. The designer should select the best alloy that matches his design as it relates to the function of the mechanical systems, the costs, and possible procurement restrictions. Nevertheless, in some cases, aluminum alloys do not guarantee an optimized structure. In particular, when an extremely light frame is needed or the design complexity of a specific structure impedes the use of traditional manufacturing technologies, traditional aluminum alloys may not be optimal. For this reason, in some specific cases, it is possible to request a waiver and adopt different materials for the main structure.

The next section will focus on new materials and new manufacturing technologies that can be used to manufacture CubeSat frames and secondary structures.

### 4.1 *New manufacturing technologies*

Different developers are focusing their attention to the use of new manufacturing technologies that guarantee more flexibility during the design process and to get an optimized solution in a reduced time. New technologies are mainly based on additive manufacturing solutions and in particular on 3-D printing. Compared to manufacturing through traditional machining, 3-D printing technology has several advantages including short periods of manufacturing, relatively high accuracy in manufacturing small parts, and low cost [12].

In general, it is possible to use additive manufacturing as a solution to manufacture not only single parts or secondary structures but also complete satellite chassis. This solution allows developers to enable new applications for CubeSats. For example, a design has been recently developed for a 3-D-printed aluminum CubeSat structure that incorporates a cold-gas propulsion system [13]. Since many CubeSat missions are more limited by volume allocation than by mass allocation, volumetric efficiency of a propulsion system is crucial. In addition, the flexibility of additive manufacturing allows the exact location and angles of the nozzles to be tailored to a specific trajectory

or set of trajectories. The use of the additive manufacturing induces a complete revolution of the design process. An overview of additive manufacturing technologies and how to identify optimal design strategies for additive manufacturing has been considered in various studies, for example, by Gaudenzi et al. in [14]. Additive manufacturing is not a complete or effective solution for all structures given that it has limitations too, but new possibilities for CubeSat structures are certainly facilitated by 3-D printing.

## 4.2 New materials

Additive manufacturing does not always mean the use of new advanced materials. The general approach is to use the capability to 3-D print traditional materials such as aluminum and/or to generate new structural solutions. Nevertheless, the new trend is to space qualify new materials typically used in additive manufacturing. Of course, not all of these materials are suitable for space applications and the restrictions in terms of outgassing and other proprieties have always to be considered. In the following, two examples of materials qualified for space applications that have been effectively used will be described.

### 4.2.1 Windform XT 2.0

In 2016 the TuPOD, the first completely 3-D-printed satellite, was launched into space. TuPOD is the acronym for TubeSat PicoSatellite Orbital Deployer, result of a cooperation between GAUSS Srl and TetonSys. Similar to the UniSat platform from GAUSS, the TUPOD was designed to be a completely autonomous satellite with the main goal of releasing two smaller satellites once on orbit. Considering its shape, it is directly analogous to a 3U CubeSat; see Fig. 3 [15].

The satellite was completely made of Windform XT 2.0, a carbon microfiber-reinforced polymer material produced by CRP Technology. The material mechanical proprieties are shown in Table 1. The technology adopted to manufacture the satellite was the selective laser sintering (SLS) method. The satellite mission was a success, and the structure performed properly during the in orbit phase, releasing the daughter satellites in the proper orbits.



**Fig. 3** TUPOD structure: X+ face to the right, Z+ face to the left.

**Table 1** Windform XT 2.0 mechanical properties.

| Mechanical properties       | WINDFORM XT 2.0 |
|-----------------------------|-----------------|
| $\rho$ (kg/m <sup>3</sup> ) | 1097            |
| $\sigma_Y$ (MPa)            | 48              |
| E (GPa)                     | 8               |

The use of rapid prototyping allows developers to have great design flexibility. For example, the main internal section of TU-POD was a unique block divided in two completely separated areas (payload and bus) avoiding the use of any secondary surfaces or components, reducing weight and optimizing the available room. While the TU-POD was the first satellite to have a structure completely 3-D printed in Windform XT 2.0, it was not the first satellite to use the Windform XT 2.0 material, as other satellites such as KySat-2 and SERPENS had already adopted it to design components or secondary structures such as camera annulus, lens cover, deployable extensions, antenna clips, and battery holders.

4.2.2 Polyether Ether Ketone (PEEK)

PEEK is a polyether ketone material under study at the European Space Agency in cooperation with the Portuguese company PIEP. It is an advanced thermoplastic with very good intrinsic properties in terms of strength, stability, and temperature resistance, with a melting point of 370°C [16]. The 3-D-printed PEEK CubeSat developed by ESA and PIEP is light, and, compared with other satellite fabrication methods, is cheap to produce. This material in addition to having all the benefits of new advanced materials such as being lightweight, cheaper, and faster to produce has the additional advantage of allowing developers to incorporate electrically conductive lines in place of the wire harness that normally connect the different CubeSat subsystems, reducing problems related to cabling and harnessing. As a next step in the use of this material, ESA is researching a space-optimized PEEK printer for initial testing on “zero-g” aircraft flights that will eventually be used by astronauts on the International Space Station.

5 Tests

The final phase before the launch is the testing phase. A section of this book (Part IV: CubeSat AITV activities) is dedicated to the description of the environmental tests that are required or suggested to be performed before launch. This section deals with the tests relevant to CubeSat structures, which should be performed at several levels: from the subsystems to the entire satellite.

In this phase, it should be emphasized that the load levels to be applied during the tests depend on the launch vehicle. Nevertheless, the developer can perform a wider variety of tests, even at higher levels than those required by the launch vehicle. These

tests and test levels can include data coming from different launch vehicles. This will allow developers to maintain an important level of flexibility during the launcher selection and to be able to move from one launch opportunity to another. This is an approach adopted by JAXA [5] that allows launches from International Space Station.

During the mission planning phase, it is important to consider the possibility of developing two nearly identical satellites: one engineering model (EM) and one flight unit (FU). Of course, budget and time might limit this approach, but this solution will allow designers to perform more invasive tests such as the qualification tests, normally performed at higher level than acceptance tests, on the EM, while testing the FU at lower levels during the acceptance tests.

## 6 CubeSat-derived form factors

Following the invention of the CubeSat standard, new small satellite form factors have been introduced. An overview of the most commonly adopted ones, along with their respective deployers and the main mission performed up to date, is provided.

### 6.1 PocketQubeSat

PocketQubeSats (initially called PocketQubs) are pico-class satellites (mass between 0.1 and 1 kg) that are envisioned to fly in constellations and perform a variety of functions from communications to Earth remote sensing and space research. The PocketQubeSat standard was first proposed by Prof. Bob Twiggs (Morehead State University) in 2009, with the idea of having a smaller and cheaper satellite than a CubeSat, and which could fit in a pocket, hence the name PocketQube. With the PocketQubeSat idea the concept of a “personal satellite” was introduced, a satellite that can be developed by a single developer interested in having its own satellite in space and able to afford the mission costs. The original concept envisioned the entire mission cost to be lower than the cost of an automobile.

The PocketQube dimensions are 1/8th of CubeSat since this miniaturized satellite has a size of 5 cm cubed. It is possible to design different units of PocketQubeSat remembering that the letter “p” indicates a unit of 5x5x5 cm. Regarding the mass, 1p weights no more than 180g. The electronic bus typically uses commercial off-the-shelf components [17]. Several PocketQubeSat dimensions have been considered in the last decade. In 2018 GAUSS Srl, TU Delft and Alba Orbital standardized the external dimensions publishing the PocketQube Standard available at [18].

#### 6.1.1 Deployer

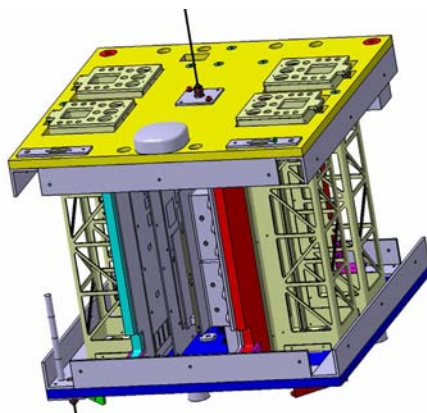
PocketQubeSats can be placed in orbit using a deployer mechanism called MRFOD (Morehead Rome Femtosatellite Orbital Deployer). The MRFOD system is the result of a joint project between the Space Science Center at Morehead State University and GAUSS (Group of Astrodynamics of University of Roma Sapienza) at University of

Rome Sapienza. The MRFOD is directly analogous to the P-POD with the main difference being that, instead of having four rails, the FOD features slotted rail systems toward the bottom of the unit. PocketQubes have extended bottom faces (tabs), which allow them to slide along the guide rails. The ejecting force is created using an extension spring and a pusher plate located toward the rear of the guide rails.

MRFOD was first tested on board the EduSat.it satellite developed by GAUSS at the School of Aerospace Engineering in Rome and funded by the Italian Space Agency. The purpose of this first mission was to test the opening mechanism without the release of satellites. The first launch of PocketQubeSats was performed using MRFOD during the UniSat-5 mission. UniSat-5 was the first satellite designed, built, and launched in 2013 by GAUSS Srl (Group of Astrodynamics for the Use of Space Systems), the company that spun out of the experience of the UniSat program at the School of Aerospace Engineering of the University of Rome Sapienza. A CAD representation of EduSat.it with the MRFODs installed is shown in Fig. 4.

### 6.1.2 Maiden mission

During the UniSat-5 flight, four PocketQubeSats were launched. The smallest one, called WREN, was produced by the German startup STADOKO UG. Even though it was a 1p PocketQubeSat, it was able to carry onboard four pulsed plasma thrusters, a three axis reaction wheel, and a color camera. Two of the other PocketQubeSats were designed at Morehead State University with the support of various partners. Both of them were designed to provide a component test bed for various spacecraft technologies, primarily among them being a deorbit system that also increased the spacecraft radar cross section. In particular Eagle-1 (aka T-LogoQube) was a 1.5p PocketQube, and Eagle-2 (aka \$50SAT) was a 2.5p PocketQube. T-LogoQube/Eagle-1 was developed by Morehead State University and Sonoma State, and \$50Sat/Eagle-2 was developed by an Amateur Radio group and Morehead State University. T-LogoQube/Eagle-1 and \$50Sat were successful. The \$50 sat worked over 2 years, and T-LogoQube worked for about 6 months. In the same mission another 2.5p



**Fig. 4** EduSat.it with the MRFODs installed.

PocketQubeSat was boarded, called QBScout. Developed by University of Maryland BC, LACO Lab, it boarded a fine sun sensor. One of its main goals was the dynamic measurement of altitude.

## 6.2 TubeSat

The TubeSat is a satellite standard proposed by the US company Interorbital System (IOS). The purpose of IOS is to provide a low-cost alternative to CubeSats, since TubeSat has three-quarters of the mass (0.75 kg or 1.65 lb) and volume of a CubeSat [13] and can be launched using the ultra-low-cost NEPTUNE orbital launch vehicle developed and built by Interorbital Systems. Nevertheless, up to now, no TubeSats have flown in orbit using this system. The only two TubeSats launched to date (Tancredo I, OSNsat) have used the TuPOD previously described in [Section 3.2.1](#).

Since its weight is less than 0.75 kg, it can be classified as a picosatellite. The TubeSat envelope is almost a cylinder of 8.94 cm of external diameter and 12.57-cm height. The TubeSat Kit sold by IOS has a hexadecagon shape, and it is assembled from a set of printed circuit boards. Nevertheless, some developers (i.e., Open Space Network) have decided to build their own structure and subsystems, maintaining the external dimensions and the weight defined by IOS.

### 6.2.1 First TubeSat mission

The first TubeSat missions were performed using the TuPOD system. In particular the TuPOD was carried on the JAXA spacecraft “KOUNOTORI-6” launched on December 9, 2016, on a mission to resupply the International Space Station. The TuPOD was deployed from the Japanese Experiment Module “Kibo” on January 16, 2016. In accordance with ISS safety regulations, on January 19, 2017, at 23:30 UTC, the TuPOD successfully dispensed the two TubeSats Tancredo I and OSNSAT. One hour later, at around 00:30 UTC, the satellite contacted the GAUSS ground station in Rome indicating the successful deployment of the TubeSats [19].

The Tancredo 1 TubeSat is the first satellite of the Ubatuba Sat Project, designed by the primary school Presidente Tancredo de Almeida Neves, located in Ubatuba, Brazil. The satellite was developed using the IOS commercial kit that was significantly modified and improved by the project team. Tancredo 1 had an onboard voice recorder IC as payload that transmitted a message chosen by a contest among the students.

OSNsat was the first satellite developed by the US company Open Space Network (OSN). In this case the developer decided not to adopt the IOS kit but to design its own platform using additive manufacturing technologies and to use part of the electronics already tested on 50\$Sat. Because of its short life, the satellite was equipped only with primary batteries and did not use solar cells. The objective of OSNSat mission was to test new technologies for satellite communications.

### 6.3 *ThinSat*

ThinSat is a relatively new standard, invented in 2016 by Twiggs Space Lab, LLC (TSL), and Morehead State University. ThinSat has a mass of 250 g and is sold by TSL in kits equipped with flexible solar cells on both sides and a GlobalStar Antenna. The idea is very similar to the PocketQubeSats, but the use of a form factor of  $4.5'' \times 4.5'' \times 5/8''$  allows better accommodation of batteries and other internal components. What is completely new on ThinSats is that seven satellites are tethered one to the other. This concept allows a simplification of the documentation needed for the launch and in orbit operation; for example, one satellite needs to be licensed as opposed to seven individual ThinSats. In addition, the form factor allows developers to easily accommodate 21 ThinSat (three string of seven satellites) in a 3U CubeSat System Deployer (CSD).

#### 6.3.1 *Maiden mission and launch opportunities*

Virginia Space and program participants launched 60 ThinSats on April 17, 2019, as a secondary payload on Northrop Grumman's Antares rocket from Pad 0A of the Mid-Atlantic Regional Spaceport located at NASA Wallops Flight Facility. Forty-nine of the ThinSats carrying standard and custom payloads successfully transmitted data via the orbiting Globalstar constellation system. Virginia Commercial Space Flight Authority (VCSFA) and TSL have the opportunity to launch up to 96 PocketQubeSat, 84 ThinSat, or other picosatellite equivalents on four (4) 3U containerized satellite dispensers on each of the Antares resupply missions through June 30, 2021. Additional capacity may be added to meet future demand.

### 6.4 *ChipSat*

Developed at Cornell University in Ithaca, NY, USA, SpriteSats or "ChipSats" are femtosatellites of  $3.2 \times 3.2$  cm with a thickness of a few millimeter. In the size of a couple of postage stamps, they are able to board solar cells, a radio transceiver, and a microcontroller. The idea is to deploy the ChipSats using a 3U CubeSat. In particular, Cornell University developed in 2011 the KickSat satellite with the aim to deploy in orbit 104 SpriteSats. KickSat was launched with the Dragon cargo capsule on April 18, 2014 (19:25:22 UTC) from Cape Canaveral, FL, but unfortunately, the system was not able to command the Sprite deployment in time. The 104 Sprite satellites were supposed to be released on May 4, 2014, but a reset of the timer by the onboard "watchdog" microcontroller on April 30, 2014, did not allow the release of the satellites before the mothership KickSat reentered and burned up in the atmosphere (May 14, 2014).

Subsequently the team attempted a second launch of KickSat-2. In March 2019 105 ChipSat miniprobes were successfully contacted, 1 day after deploying from their KickSat-2 carrier spacecraft. The success of the ChipSats has implications for an accelerated "democratization of space."

## 7 Conclusions

In this chapter an overview of best practices during the structural system design and manufacturing has been presented. In particular, attention has been focused on the definition of the requirements and how they affect the overall satellite design. Some aspects are strictly related to the mission itself and cannot be analyzed in detail, but the developer can use this chapter to analyze their particular case and find the best solution to be adopted.

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